Fundamental groupoid of a homotopy pushout via the naive van Kampen code

The Univalent Foundations Program

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Editorial note. This counts the book labellednaive van Kampen theorem only. The later refined/classical formulation is not an additional count.

Editorial note. The original proof compresses the symmetric endpoint cases and the final word induction; these are retained as author-level routine omissions, not replaced by generated arguments.

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Allowed standard prerequisites (editorial).

- Explicit univalent foundations boundary. Univalence, function extensionality, higher-inductive pushouts, set quotients,0-truncation and path groupoid operations. The word-code construction and its compatibility are part of the original retained proof.

Source reference. Original section 8.1: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1694–1694. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: section.

Source reference. Original exercise 9.11: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1399–1399. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: ex.

Source reference. Original section 2.12: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1625–1625. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: section.

Source reference. Original section 2.13: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1626–1626. Original book reference to explicitly allowed

type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: section.

Source reference. Original section 8.7.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1611–1611. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: subsection.

Source reference. Original section 6.8: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1620–1620. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: section.

Source reference. Original section 6.10: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1704–1704. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: section.

8.7 The van Kampen theorem

The van Kampen theorem calculates the fundamental group π_1 of a (homotopy) pushout of spaces. It is traditionally stated for a topological space X which is the union of two open subspaces U and V, but in homotopy-theoretic terms this is just a convenient way of ensuring that X is the pushout of U and V over their intersection. Thus, we will prove a version of the van Kampen theorem for arbitrary pushouts.

In this section we will describe a proof of the van Kampen theorem which uses the same encode-decode method that we used for $\pi_1(\mathbb{S}^1)$ in original section 8.1. There is also a more homotopy-theoretic approach; see original exercise 9.11.

We need a more refined version of the encode-decode method. In original section 8.1 (as well as in original sections 2.12 and 2.13) we used it to characterize the path space of a (higher) inductive type W — deriving as a consequence a characterization of the loop space $\Omega(W)$, and thereby also of its 0-truncation $\pi_1(W)$. In the van Kampen theorem, our goal is only to characterize the fundamental group $\pi_1(W)$, and we do not have any explicit description of the loop spaces or the path spaces to use.

It turns out that we can use the same technique directly for a truncated version of the path fibration, thereby characterizing not only the fundamental *group* $\pi_1(W)$, but also the whole fundamental *groupoid*. Specifically, for a type X, write $\Pi_1 X : X \to X \to \mathcal{U}$ for the 0-truncation of its identity type, i.e.

$\Pi_1 X(x,y) :\equiv \|x = y\|_0$. Note that we have induced groupoid operations

$$\begin{aligned}(- \cdot -) &: \Pi_1 X(x,y) \to \Pi_1 X(y,z) \to \Pi_1 X(x,z)\\ (-)^{-1} &: \Pi_1 X(x,y) \to \Pi_1 X(y,x)\\ \mathsf{refl}_x &: \Pi_1 X(x,x)\\ \mathsf{ap}_f &: \Pi_1 X(x,y) \to \Pi_1 Y(fx,fy)\end{aligned}$$

for which we use the same notation as the corresponding operations on paths.

8.7.1 Naive van Kampen

We begin with a "naive" version of the van Kampen theorem, which is useful but not quite as useful as the classical version. In original section 8.7.2 we will improve it to a more useful version.

Given types A, B, C and functions $f : A \to B$ and $g : A \to C$, let P be their pushout $B \sqcup^A C$. As we saw in original section 6.8, P is the higher inductive type generated by

- $i : B \to P$,
- $j : C \to P$, and
- for all $x : A$, a path $kx : ifx = jgx$.

Define $\mathsf{code} : P \to P \to \mathcal{U}$ by double induction on P as follows.

- $\mathsf{code}(ib, ib')$ is a set-quotient (see original section 6.10) of the type of sequences
$$(b, p_0, x_1, q_1, y_1, p_1, x_2, q_2, y_2, p_2, \dots, y_n, p_n, b')$$
where
  - $n : \mathbb{N}$
  - $x_k : A$ and $y_k : A$ for $0 < k \le n$
  - $p_0 : \Pi_1 B(b, fx_1)$ and $p_n : \Pi_1 B(fy_n, b')$ for $n > 0$, and $p_0 : \Pi_1 B(b, b')$ for $n = 0$
  - $p_k : \Pi_1 B(fy_k, fx_{k+1})$ for $1 \le k < n$
  - $q_k : \Pi_1 C(gx_k, gy_k)$ for $1 \le k \le n$

  The quotient is generated by the following equalities:
$$\begin{aligned}(\dots, q_k, y_k, \mathsf{refl}_{fy_k}, y_k, q_{k+1}, \dots) &= (\dots, q_k \cdot q_{k+1}, \dots)\\ (\dots, p_k, x_k, \mathsf{refl}_{gx_k}, x_k, p_{k+1}, \dots) &= (\dots, p_k \cdot p_{k+1}, \dots)\end{aligned}$$
  (see Remark 8.7.3 below). We leave it to the reader to define this type of sequences precisely as an inductive type.
- $\mathsf{code}(jc, jc')$ is identical, with the roles of B and C reversed. We likewise notationally reverse the roles of x and y, and of p and q.
- $\mathsf{code}(ib, jc)$ and $\mathsf{code}(jc, ib)$ are similar, with the parity changed so that they start in one type and end in the other.

- For $a : A$ and $b : B$, we require an equivalence

$$\mathsf{code}(ib, ifa) \simeq \mathsf{code}(ib, jga). \tag{8.7.1}$$

  We define this to consist of the two functions defined on sequences by

$$\begin{aligned}(\dots, y_n, p_n, fa) &\mapsto (\dots, y_n, p_n, a, \mathsf{refl}_{ga}, ga),\\ (\dots, x_n, p_n, a, \mathsf{refl}_{fa}, fa) &\mapsto (\dots, x_n, p_n, ga).\end{aligned}$$

  Both of these functions are easily seen to respect the equivalence relations, and hence to define functions on the types of codes. The left-to-right-to-left composite is

$$(\dots, y_n, p_n, fa) \mapsto (\dots, y_n, p_n, a, \mathsf{refl}_{ga}, a, \mathsf{refl}_{fa}, fa)$$

  which is equal to the identity by a generating equality of the quotient. The other composite is analogous. Thus we have defined an equivalence (8.7.1).
- Similarly, we require equivalences

$$\begin{aligned}\mathsf{code}(jc, ifa) &\simeq \mathsf{code}(jc, jga)\\ \mathsf{code}(ifa, ib) &\simeq (jga, ib)\\ \mathsf{code}(ifa, jc) &\simeq (jga, jc)\end{aligned}$$

  all of which are defined in exactly the same way (the second two by adding reflexivity terms on the beginning rather than the end).
- Finally, we need to know that for $a, a' : A$, the following diagram commutes:

$$\begin{array}{ccc}\mathsf{code}(ifa, ifa') & \longrightarrow & \mathsf{code}(ifa, jga')\\ \downarrow & & \downarrow\\ \mathsf{code}(jga, ifa') & \longrightarrow & \mathsf{code}(jga, jga')\end{array} \tag{8.7.2}$$

  This amounts to saying that if we add something to the beginning and then something to the end of a sequence, we might as well have done it in the other order.

Remark 8.7.3. One might expect to see in the definition of code some additional generating equations for the set-quotient, such as

$$\begin{aligned}(\dots, p_{k-1} \cdot fw, x'_k, q_k, \dots) &= (\dots, p_{k-1}, x_k, gw \cdot q_k, \dots) \quad (\text{for } w : \Pi_1 A(x_k, x'_k))\\ (\dots, q_k \cdot gw, y'_k, p_k, \dots) &= (\dots, q_k, y_k, fw \cdot p_k, \dots). \quad (\text{for } w : \Pi_1 A(y_k, y'_k))\end{aligned}$$

However, these are not necessary! In fact, they follow automatically by path induction on w. This is the main difference between the "naive" van Kampen theorem and the more refined one we will consider in the next subsection.

Continuing on, we can characterize transporting in the fibration code:

- For $p : b =_B b'$ and $u : P$, we have

$$\mathsf{transport}^{b \mapsto \mathsf{code}(u,ib)}(p, (\dots, y_n, p_n, b)) = (\dots, y_n, p_n \cdot p, b').$$

- For $q : c =_C c'$ and $u : P$, we have

$$\mathsf{transport}^{c \mapsto \mathsf{code}(u,jc)}(q, (\dots, x_n, q_n, c)) = (\dots, x_n, q_n \cdot q, c').$$

Here we are abusing notation by using the same name for a path in X and its image in $\Pi_1 X$. Note that transport in $\Pi_1 X$ is also given by concatenation with (the image of) a path. From this we can prove the above statements by induction on u. We also have:

- For $a : A$ and $u : P$,

$$\mathsf{transport}^{v \mapsto \mathsf{code}(u,v)}(ha, (\dots, y_n, p_n, fa)) = (\dots, y_n, p_n, a, \mathsf{refl}_{ga}, ga).$$

This follows essentially from the definition of code.

We also construct a function

$$r : \prod_{u:P} \mathsf{code}(u, u)$$

by induction on u as follows:

$$rib :\equiv (b, \mathsf{refl}_b, b)$$
$$rjc :\equiv (c, \mathsf{refl}_c, c)$$

and for rka we take the composite equality

$$\begin{aligned}(ka, ka)_*(fa, \mathsf{refl}_{fa}, fa) &= (ga, \mathsf{refl}_{ga}, a, \mathsf{refl}_{fa}, a, \mathsf{refl}_{ga}, ga) \\ &= (ga, \mathsf{refl}_{ga}, ga)\end{aligned}$$

where the first equality is by the observation above about transporting in code, and the second is an instance of the set quotient relation used to define code.

We will now prove:

Theorem 8.7.4 (Naive van Kampen theorem)**.** *For all $u, v : P$ there is an equivalence*

$$\Pi_1 P(u, v) \simeq \mathsf{code}(u, v).$$

Proof. To define a function

$$\mathsf{encode} : \Pi_1 P(u, v) \to \mathsf{code}(u, v)$$

it suffices to define a function $(u =_P v) \to \mathsf{code}(u, v)$, since $\mathsf{code}(u, v)$ is a set. We do this by transport:

$$\mathsf{encode}(p) :\equiv \mathsf{transport}^{v \mapsto \mathsf{code}(u,v)}(p, r(u)).$$

Now to define

$$\mathsf{decode} : \mathsf{code}(u, v) \to \Pi_1 P(u, v)$$

we proceed as usual by induction on $u, v : P$. In each case for u and v, we apply i or j to all the equalities p_k and q_k as appropriate and concatenate the results in P, using h to identify the endpoints. For instance, when $u \equiv ib$ and $v \equiv ib'$, we define

$$\mathsf{decode}(b, p_0, x_1, q_1, y_1, p_1, \dots, y_n, p_n, b') :\equiv (p_0) \cdot h(x_1) \cdot j(q_1) \cdot h(y_1)^{-1} \cdot i(p_1) \cdot \dots \cdot h(y_n)^{-1} \cdot i(p_n). \tag{8.7.5}$$

This respects the set-quotient equivalence relation and the equivalences such as (8.7.1), since $h : fi \sim gj$ is natural and f and g are functorial.

As usual, to show that the composite

$$\Pi_1 P(u,v) \xrightarrow{\mathsf{encode}} \mathsf{code}(u,v) \xrightarrow{\mathsf{decode}} \Pi_1 P(u,v)$$

is the identity, we first peel off the 0-truncation (since the codomain is a set) and then apply path induction. The input refl_u goes to ru, which then goes back to refl_u (applying a further induction on u to decompose $\mathsf{decode}(ru)$).

Finally, consider the composite

$$\mathsf{code}(u,v) \xrightarrow{\mathsf{decode}} \Pi_1 P(u,v) \xrightarrow{\mathsf{encode}} \mathsf{code}(u,v).$$

We proceed by induction on $u, v : P$. When $u \equiv ib$ and $v \equiv ib'$, this composite is

$$\begin{aligned}
(b, p_0, x_1, q_1, y_1, p_1, \dots, y_n, p_n, b') &\mapsto \left(ip_0 \cdot hx_1 \cdot jq_1 \cdot hy_1{}^{-1} \cdot ip_1 \cdot \dots \cdot hy_n{}^{-1} \cdot ip_n\right)_* (rib) \\
&= (ip_n)_* \cdots (jq_1)_* (hx_1)_* (ip_0)_* (b, \mathsf{refl}_b, b) \\
&= (ip_n)_* \cdots (jq_1)_* (hx_1)_* (b, p_0, if x_1) \\
&= (ip_n)_* \cdots (jq_1)_* (b, p_0, x_1, \mathsf{refl}_{gx_1}, jgx_1) \\
&= (ip_n)_* \cdots (b, p_0, x_1, q_1, jgy_1) \\
&= \quad \vdots \\
&= (b, p_0, x_1, q_1, y_1, p_1, \dots, y_n, p_n, b').
\end{aligned}$$

i.e., the identity function. (To be precise, there is an implicit inductive argument needed here.) The other three point cases are analogous, and the path cases are trivial since all the types are sets. □